Polynomials and Linear Factors

Review: Write the polynomial in standard form.

$x^3 + 2 - 3x - 2x^2$ What is the degree?

** Remember: In standard form, the _____ degree goes first.

Example 1: Write the following polynomials in factored form. (Always look to see if you could factor out a # or variable first)

A) $f(x) = 3x^3 - 18x^2 + 24x$

B) $f(x) = -2x^2 + 8x - 8$

The _____ of a function is where the graph crosses the ___-axis.

AKA: zero = _____ = x-intercept

To find the zeros

1. Write the equation in factored form.
2. Set each factor equal to zero and solve.

A _____ zero is called a multiple zero.

The _____ is the number of times that zero occurs.

Example 3: Find the zeros of the following functions. Then graph the function with the help of your graphing calc to find the max/mins

A) $f(x) = (x-1)(x+1)(x+3)^2$

B) $y = x^2 + 4x - 5$

Example 4: Write a polynomial function in standard form with zeros at

A) $2, -3, and\ 0$.

B) $1(mult\ 2) and -4$